

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tool

- Anti-seize compound, Part Number 3824879, or equivalent

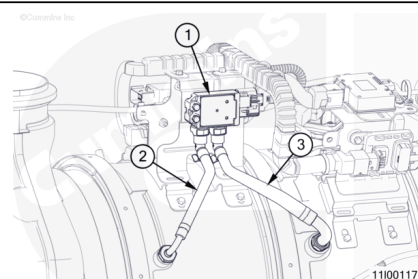
Additional Service Items

- No additional service items required.

General Information

⚠ WARNING ⚠

During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the engine has been shut off. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.



⚠ WARNING ⚠

The material captured in a partial flow diesel particulate filter and/or a diesel particulate filter may contain elevated concentrations of metals. Primarily zinc, molybdenum, and possibly polynuclear aromatic hydrocarbons, that may be regulated. These materials must be characterized, handled, and disposed of according to applicable local regulations. In addition, due to the presence of the above-listed chemicals and other potentially toxic components such as oxides of

calcium, zinc, phosphorous, silicon, sulfur, and iron, exhaust filter maintenance must be completed only by appropriately trained personnel.

The aftertreatment diesel particulate filter (DPF) differential pressure sensor tubes connect the aftertreatment diesel particulate filter differential pressure sensor (1) to the ports on the aftertreatment system. There are two aftertreatment diesel particulate filter differential pressure sensor tubes. One tube (2) connects to the aftertreatment system upstream of the aftertreatment diesel particulate filter and the other tube (3) connects downstream of the aftertreatment diesel particulate filter.

Due to the number of aftertreatment applications, this procedure is generic. **Not** all illustrations within the procedure will represent all applications.

Preparatory Steps

⚠ WARNING ⚠

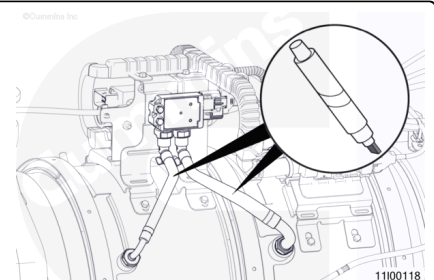
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

Remove

⚠ WARNING ⚠

During regeneration, exhaust gas temperature can reach 800 °C [1500°F], and exhaust system surface temperature can exceed 700 °C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the engine has been shut off. To avoid the risk of fire, property damage, burns, or other serious personal injury, allow the exhaust



system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

⚠ CAUTION ⚠

The aftertreatment DPF differential pressure sensor will not operate properly if the differential pressure sensor tubes are not connected to the correct port. Mark the differential pressure sensor tube connection port locations before disconnecting.

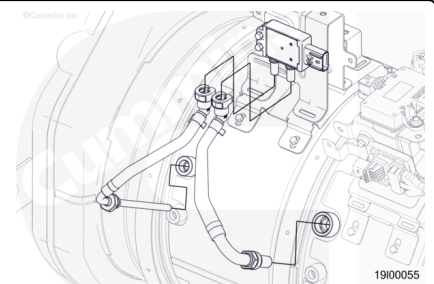
Note : The mounting location of the aftertreatment DPF differential pressure sensor varies with exhaust aftertreatment configuration.

Disconnect the differential pressure sensor tubes from the sensor by releasing the quick connect locking tab and pulling the tubes off the sensor. Refer to Procedure 019-443 in Section 19.

If P-clips or tube clamps are used to hold the aftertreatment DPF differential pressure sensor tubes on the aftertreatment system, mark their locations prior to removal.

Remove the P-clips or tube clamp mounting capscrews, if necessary.

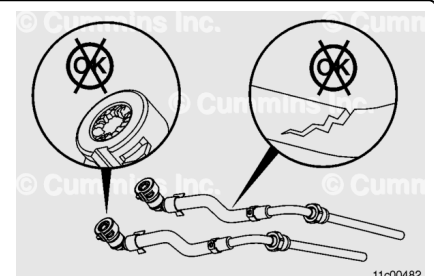
Loosen sensor tube port fittings on the aftertreatment DPF canister and remove the sensor tube.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

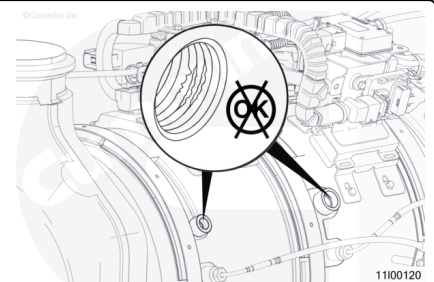
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Inspect the tube for the following:

- Clogging: If the tube is partially clogged, saturate the inside of the tube with a mineral-based solvent or equivalent safety solvent. Dry the tube with compressed air. If the tube is fully clogged or otherwise unable to be cleaned, the tube **must** be replaced.
- Cracking: If the tube is cracked, it **must** be replaced.
- Sensor tube port fittings damage: Replace the tube if damage is found.

Inspect the inside of the threaded bosses on the aftertreatment canister for the following:

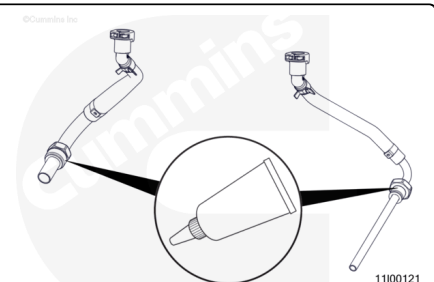
- Clean debris from the inside of the threaded bosses, while being careful to **not** damage the threads.
- Thread damage. Repair the thread damage, if necessary.



Install

⚠ CAUTION ⚠

The aftertreatment DPF differential pressure sensor will not operate properly if the differential pressure sensor tubes are not connected to the correct port. Install the differential pressure sensor tubes as noted during disassembly.



⚠ CAUTION ⚠

Do not allow the anti-seize compound to get inside the aftertreatment diesel particulate filter differential pressure sensor tubes. This can cause a blockage.

Apply a coating of anti-seize compound, Part Number 3824879, or equivalent, to the threads on the aftertreatment DPF filter differential pressure sensor tubes prior to assembly.

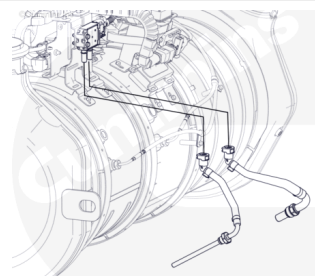
Measurements

	celsius	fahrenheit
Anti-seize Compound Minimum Temperature Range:	870	1600

Note : The aftertreatment system must be installed so the aftertreatment DPF differential pressure sensor tubes slope downward to drain condensation away from the differential pressure sensor.

Install the aftertreatment DPF differential pressure sensor tubes on the aftertreatment DPF canister. Do **not** tighten the aftertreatment DPF differential pressure sensor tube port fittings at this time. Refer to Procedure 019-443 in Section 19.

Connect the aftertreatment DPF differential pressure sensor tubes to the aftertreatment DPF differential pressure sensor. Refer to Procedure 019-443 in Section 19.



11100119

Note : Verify the aftertreatment DPF differential pressure sensor tubes are not making contact with each other or any other components prior to tightening the aftertreatment DPF differential pressure sensor tube port fittings.

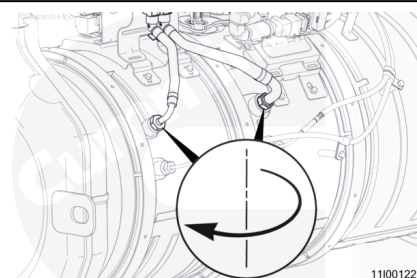
Tighten the upstream and downstream aftertreatment DPF differential pressure sensor tube port fittings.

Torque Value: 31 n•m [23 ft-lb]

Note : Vertical aftertreatment configurations require a support clip to be installed to support the aftertreatment DPF differential pressure sensor tube. For DPF mounted aftertreatment differential pressure sensors, make sure a P-clips is installed, if equipped, to support the aftertreatment DPF differential pressure sensor.

Install the P-clip or tube clamp that holds the aftertreatment DPF differential pressure sensor tube on the aftertreatment system, if necessary.

Torque Value: 11 n•m [98 in-lb]



11100122

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Verify the continuously downward slope of the aftertreatment DPF differential pressure sensor tubes from sensor connectors to sensor port fittings. Make adjustments, if required.
- Connect the batteries. See equipment manufacturer service information.
- Use INSITE™ electronic service tool to check for fault codes.
- Check for fault codes and exhaust leaks.